

1. Identification Phase

The identification phase is the initial stage of the project where the main problem and objectives of the system are clearly defined. In many organizations using ServiceNow, users often submit forms without entering all the required information. This leads to incomplete records in the system, which can cause difficulties for support teams when they try to understand or resolve issues. Because of this problem, it becomes necessary to identify a solution that ensures users provide all the essential details before submitting a form.

During this phase, the need for implementing mandatory fields in ServiceNow forms was identified. Mandatory fields ensure that users cannot submit a request or incident without filling important information such as category, description, or priority. Without these details, the support team may not be able to process the request properly. Therefore, enforcing mandatory fields helps improve the accuracy and reliability of the data stored in the system.

Another important aspect identified in this phase is the need to manage configuration changes effectively. When administrators create UI Policies or modify system configurations, those changes must be safely transferred from one ServiceNow instance to another. This requirement led to the use of Update Sets, which allow administrators to capture configuration changes and migrate them between development, testing, and production environments.

The identification phase also involves defining the scope of the project and selecting the appropriate tools and technologies required for implementation. In this project, the main tools include the ServiceNow platform, UI Policies for controlling field behaviour, and Update Sets for managing configuration migration. Identifying these tools at the beginning of the project helps in planning the development process efficiently.

Overall, the identification phase plays a crucial role in understanding the existing problems in the system and determining the best possible solution. By clearly defining the project objectives and requirements at this stage, the development process

becomes more organized and focused, leading to the successful implementation of mandatory fields using UI Policies and migration of configurations using Update Sets.

2. Requirement Analysis

2.1 Overview of Requirement Analysis

Requirement analysis is an important phase in the development of any project. In this phase, all the necessary requirements of the system are identified, studied, and documented before the actual development process begins. For this project, the requirements focus on implementing mandatory fields in forms and managing configuration changes effectively in the ServiceNow platform.

The main goal of this phase is to understand what the system needs to achieve and how the required functionalities will be implemented. By analysing the requirements clearly, developers can avoid confusion during the development process and ensure that the system meets the intended objectives.

2.2 Functional Requirements

Functional requirements describe the specific operations and features that the system must perform. In this project, the main functional requirement is to ensure that certain fields in the ServiceNow forms become mandatory based on predefined conditions. This ensures that users provide all necessary information before submitting requests or incidents.

Another important functional requirement is the ability to configure UI Policies that dynamically control the behaviour of form fields. These UI Policies allow administrators to enforce mandatory fields, make fields read-only, or hide fields depending on the conditions defined.

The system should also support the use of Update Sets to capture configuration changes. This allows administrators to migrate the changes from one ServiceNow instance to another instance without manually repeating the configuration process.

2.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance expectations of the system. One of the important non-functional requirements of this project is system reliability. The system must function correctly without errors while enforcing mandatory fields in forms.

Another requirement is usability. The configuration process for UI Policies should be simple and easy for administrators to manage. ServiceNow provides a user-friendly interface that allows administrators to configure UI Policies without writing complex scripts.

Security is also an important requirement. Only authorized users such as administrators should be allowed to create or modify UI Policies and Update Sets. This helps maintain the integrity and stability of the system.

2.4 Hardware and Software Requirements

The successful implementation of this project requires certain hardware and software resources. The hardware requirements include a computer system with sufficient processing capability and memory to run a web browser and access the ServiceNow platform. A stable internet connection is also necessary because ServiceNow operates as a cloud-based service.

The software requirements include a modern web browser such as Google Chrome, Microsoft Edge, or Mozilla Firefox. Additionally, access to a ServiceNow developer instance is required to configure UI Policies, create Update Sets, and test the system functionality.

2.5 Feasibility Analysis

Feasibility analysis helps determine whether the project can be successfully implemented with the available resources and technology. In this project, the technical

feasibility is high because ServiceNow already provides built-in features such as UI Policies and Update Sets for implementing the required functionality.

The project is also economically feasible because it does not require additional expensive software or hardware. Most of the configuration is performed within the ServiceNow platform itself.

Operational feasibility is also ensured because administrators and users can easily understand and use the system without extensive training. This makes the solution practical and efficient for organizations that rely on ServiceNow for managing their workflows.

3. Project Planning

3.1 Overview of Project Planning

Project planning is an important phase in which the overall strategy and workflow of the project are defined. In this stage, the tasks required to complete the project are organized in a systematic manner. Proper planning helps in identifying the sequence of activities and ensures that each stage of the project is completed efficiently.

For this project implemented in ServiceNow, planning plays a key role in organizing the development process such as identifying mandatory fields, configuring UI Policies, and migrating configurations using Update Sets. By planning the activities clearly, the development process becomes more structured and manageable.

3.2 Project Scope and Objectives

During the planning phase, the scope of the project is clearly defined. The project mainly focuses on enforcing mandatory fields in ServiceNow forms using UI Policies and managing configuration migration using Update Sets. The scope includes identifying the fields that need validation, configuring UI Policies to control form behaviour, and testing the implemented solution.

The objectives of the project are also determined during this phase. The primary objective is to improve data accuracy by ensuring that users provide all required information before submitting forms. Another objective is to simplify configuration management by using Update Sets to migrate changes between different ServiceNow instances.

3.3 Project Timeline and Task Allocation

Project planning also involves creating a timeline that outlines when each task will be completed. The project is divided into several stages such as requirement analysis, system design, development, testing, and final demonstration. Each stage is assigned a specific time frame to ensure that the project progresses smoothly.

Task allocation is another important aspect of planning. The responsibilities for different tasks such as configuring UI Policies, managing Update Sets, and testing the system are clearly defined. This ensures that each activity is completed efficiently and within the planned schedule.

3.4 Resource Planning

Resource planning involves identifying the tools and resources required for the successful completion of the project. The main resource required for this project is access to the ServiceNow platform. A developer instance of ServiceNow is used to create UI Policies, configure forms, and capture changes using Update Sets.

Other resources include a computer system, a web browser, and a stable internet connection to access the cloud-based platform. Proper resource planning ensures that all necessary tools are available before the development phase begins.

3.5 Risk Management

Risk management is also considered during the project planning phase. Potential risks such as configuration errors, incomplete Update Sets, or system compatibility issues are identified. By recognizing these risks early, preventive measures can be implemented to reduce their impact on the project.

For example, thorough testing is planned to ensure that UI Policies work correctly and that Update Sets capture all configuration changes properly. Effective risk management helps in completing the project successfully without major issues.

4. Project Design

4.1 Overview of System Design

The project design phase focuses on creating the structure and workflow of the system before the development process begins. In this stage, the overall design of the solution is planned so that the system can function effectively and meet the project objectives.

For this project, the design mainly focuses on configuring UI Policies to enforce mandatory fields in ServiceNow forms and using Update Sets to manage configuration changes. The design phase ensures that the form structure, field validation rules, and configuration management processes are clearly defined before implementation.

4.2 Form Design and Field Structure

Form design is an important part of the project because it defines how users interact with the system. The forms in ServiceNow are designed to collect necessary information from users while submitting requests or incidents.

During this phase, important fields such as category, description, priority, and assignment group are identified. These fields contain critical information required for processing requests. The design ensures that these fields become mandatory when specific conditions are satisfied.

By structuring the form carefully, the system ensures that users provide complete and relevant information before submitting a request.

4.3 UI Policy Design

UI Policies are designed to control the behaviour of form fields dynamically. In this project, UI Policies are configured to enforce mandatory fields based on certain conditions. For example, when a user selects a specific category, additional fields may automatically become mandatory.

The UI Policy design includes defining the condition that triggers the policy and specifying the actions that will be applied to the fields. These actions may include making fields mandatory, read-only, or visible depending on the requirement.

This design approach improves usability and ensures that only relevant information is required from users.

4.4 Update Set Structure

Another important part of the design phase is planning how configuration changes will be captured and migrated. Update Sets are used to record all changes made to the system configuration.

The design ensures that all UI Policies and form configurations are included in the Update Set so that they can be transferred to another ServiceNow instance without any issues. This structured approach helps maintain consistency across different environments such as development, testing, and production.

5. Project Development

5.1 Development Environment Setup

The development phase begins with setting up the required environment. A ServiceNow developer instance is used for implementing the project. The developer logs into the platform using a web browser and accesses the system configuration modules.

The required tables and forms are identified where mandatory field enforcement needs to be implemented. This setup ensures that the development process can proceed smoothly.

5.2 Creation of UI Policies

The next step in development is creating UI Policies. The administrator navigates to the UI Policy configuration section in ServiceNow and creates a new policy for the selected form.

During this process, the condition for the UI Policy is defined. For example, a field may become mandatory when a user selects a specific category or priority. After defining the condition, UI Policy actions are created to enforce the mandatory fields.

This step ensures that the form behaviour changes dynamically based on user input.

5.3 Implementation of Mandatory Fields

Once the UI Policy is created, the mandatory fields are configured within the policy actions. These actions specify which fields must be filled before the form can be submitted.

When users interact with the form, the system automatically checks the defined conditions and applies the mandatory field rules. If the required fields are not filled, the system prevents the user from submitting the form.

This implementation improves data quality and ensures that all necessary information is collected.

5.4 Capturing Changes Using Update Sets

After completing the configuration, all changes made during development are captured in an Update Set. Update Sets track configuration changes such as UI Policies, form modifications, and field settings.

The Update Set acts as a container that stores all the modifications performed during the development process. This allows administrators to migrate the configuration to another ServiceNow instance when needed.

5.5 Testing During Development

Testing is an important part of the development phase. After configuring the UI Policies, the system is tested to ensure that the mandatory fields behave correctly.

Different scenarios are tested to verify that the fields become mandatory only when the specified conditions are satisfied. If any issues are found, the configuration is modified and tested again until the system functions properly.

6. Project Demonstration

6.1 Overview of Demonstration

The project demonstration phase involves presenting the completed system and explaining how the implemented features work. This phase is important because it

shows the practical functionality of the project and verifies that the system meets the project objectives.

During the demonstration, the working of the ServiceNow platform is explained along with the UI Policies created for enforcing mandatory fields.

6.2 Demonstration of Form Behaviour

In this stage, the ServiceNow form is opened and users interact with the fields. When the user selects certain options such as category or priority, the UI Policy automatically triggers and specific fields become mandatory.

The demonstration shows how the system prevents users from submitting the form if the mandatory fields are not filled. This clearly illustrates how the UI Policy improves data accuracy.

6.3 Demonstration of Update Set Migration

Another important part of the demonstration is showing how configuration changes are migrated using Update Sets. The Update Set created during development is retrieved and prepared for migration.

The Update Set is then imported into another ServiceNow instance, and the configuration changes are applied. This process demonstrates how administrators can easily move configurations between environments without repeating the setup process.

6.4 Output and Result Explanation

Finally, the results of the project are explained. The demonstration highlights how the system ensures that users provide complete information before submitting forms. It also shows how Update Sets help maintain consistency between different ServiceNow environments.

The output confirms that the implemented solution works successfully and fulfills the project objectives. The demonstration provides a clear understanding of the project functionality and its practical benefits in real-world scenarios.

7. System Architecture

7.1 Overview of System Architecture

System architecture describes the overall structure of the system and how different components interact with each other. In this project, the ServiceNow platform acts as the central environment where all forms, UI Policies, and configurations are managed. Users interact with the system through a web browser interface where they can create requests, fill forms, and submit information.

The architecture ensures that the user interface, form fields, UI Policies, and configuration management components work together efficiently. Whenever a user opens a form and enters information, the system automatically checks the conditions defined in the UI Policy. Based on these conditions, specific fields become mandatory or visible to the user.

7.2 Components of the System

The system architecture mainly consists of several components that work together to perform the required operations.

User Interface:

This is the front-end environment where users interact with the ServiceNow forms. Users enter the required details and submit requests through this interface.

UI Policy Engine:

This component controls the behaviour of form fields dynamically. It evaluates the conditions defined by the administrator and applies actions such as making fields mandatory.

Database:

All the data entered by users is stored in the ServiceNow database. The database maintains records of requests, incidents, and other system information.

Update Set Manager:

This component captures and manages configuration changes made in the system. It allows administrators to migrate those changes to another instance when necessary.

7.3 Working of the Architecture

When a user opens a form in ServiceNow, the system loads all the fields and UI configurations associated with that form. As the user begins entering data, the UI Policy checks whether the defined conditions are satisfied. If the condition becomes true, certain fields automatically become mandatory.

The user must fill these mandatory fields before submitting the form. Once the form is submitted successfully, the information is stored in the database. If administrators have configured changes during development, those modifications are captured in Update Sets, which can later be migrated to another ServiceNow environment.

This architecture ensures proper data validation and smooth workflow management within the system.

8. Functional Modules

8.1 Overview of Functional Modules

The project is divided into several functional modules to simplify the design and development process. Each module performs a specific task within the system and contributes to the overall functionality of the project.

Breaking the system into modules helps in managing the development process more effectively and allows each component to perform its role independently.

8.2 User Interface Module

The user interface module allows users to interact with the ServiceNow platform. It provides forms where users can enter information related to service requests or incidents. The interface is designed to be user-friendly so that users can easily understand and complete the required fields.

This module ensures that users can submit requests quickly while the system guides them to provide all necessary information.

8.3 UI Policy Module

The UI Policy module is responsible for controlling the behaviour of form fields dynamically. This module ensures that certain fields become mandatory based on specific conditions defined by the administrator.

For example, when a user selects a particular category in the form, additional fields may become mandatory. This ensures that relevant information is collected before the form is submitted.

8.4 Update Set Module

The Update Set module manages configuration changes within the system. Whenever administrators create or modify UI Policies, those changes are captured in an Update Set.

This module allows the configuration changes to be exported and imported into other ServiceNow instances. It helps maintain consistency across development, testing, and production environments.

8.5 Testing Module

The testing module verifies whether the implemented UI Policies and configurations are functioning correctly. It ensures that mandatory fields are enforced properly and that users cannot submit forms without filling required information.

Testing also ensures that Update Sets capture all configuration changes accurately and that they can be migrated without errors.

9. Data Flow Process

9.1 Overview of Data Flow

The data flow process describes how information moves through the system from the user interface to the database. In this project, data flow begins when a user accesses a form within the ServiceNow platform.

The user enters information into the form fields and interacts with the interface to complete the request submission process.

9.2 Data Input Process

When the user starts filling the form, the system accepts the input values and monitors them using UI Policies. The UI Policy evaluates the conditions defined during configuration.

If the specified condition becomes true, additional fields may become mandatory. The system then prompts the user to complete those fields before allowing the form submission.

9.3 Data Validation Process

Before the form is submitted, the system performs validation checks to ensure that all mandatory fields are completed. If any required field is left empty, the system displays an error message and prevents the form from being submitted.

This validation process ensures that incomplete data is not stored in the system and that all necessary details are captured correctly.

9.4 Data Storage Process

Once the form is successfully validated and submitted, the information entered by the user is stored in the ServiceNow database. The stored data can later be accessed by administrators or support teams to process the request.

Configuration changes made during development are stored in Update Sets, which can later be migrated to other instances when required.

10. Conclusion

10.1 Summary of the Project

This project focuses on enforcing mandatory fields in ServiceNow forms using UI Policies and migrating configuration changes using Update Sets. The system ensures that users provide all required information before submitting requests, which improves the accuracy and completeness of the data stored in the system.

The project demonstrates how ServiceNow features can be used to automate validation processes and simplify configuration management.

10.2 Achievements of the Project

The implementation of UI Policies successfully ensures that certain fields become mandatory based on predefined conditions. This prevents users from submitting incomplete forms and helps maintain reliable records in the system.

The use of Update Sets also provides an efficient way to capture configuration changes and migrate them between different ServiceNow environments.

10.3 Final Outcome

Overall, the project improves the efficiency of form validation and configuration management within the ServiceNow platform. It highlights the importance of enforcing data validation rules and managing system changes effectively.

By using ServiceNow tools such as UI Policies and Update Sets, organizations can enhance their workflow automation and maintain better control over their IT service